American Computer Science League

2024-2025 ● Contest 3: Short Problems Solutions ● Intermediate Division

1. Boolean Algebra

$$A(\overline{B + C}) + \overline{ABC} = A\,\overline{B}\,\overline{C} + \overline{A} + \overline{B} + \overline{C}$$
$$= \overline{B}(A\,\overline{C} + 1) + \overline{A} + \overline{C}$$
$$= \overline{A} + \overline{B} + \overline{C}$$

This is FALSE when each term is 0, so $\overline{A} = 0, \overline{B} = 0, \overline{C} = 0$.
The only ordered triple would be (1,1,1).

D. (1,1,1)

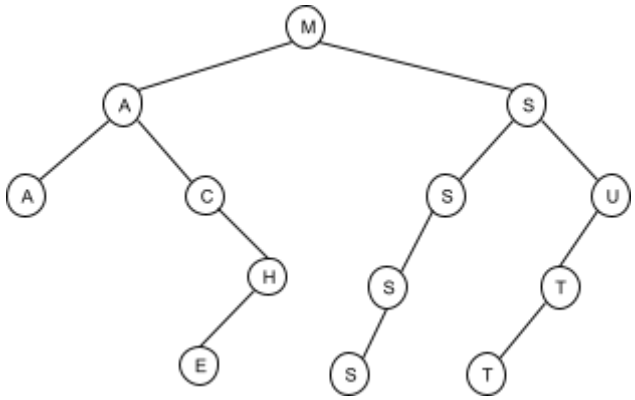
2. Boolean Algebra

$$\overline{AB}\,(\overline{A + C}) + \overline{C}$$
$$= (\overline{A} + \overline{B})\,(\overline{A} * \overline{C}) + \overline{C}$$
$$= \overline{A} * \overline{C} + \overline{A} * \overline{B} * \overline{C} + \overline{C}$$
$$= (\overline{A} + \overline{A} * \overline{B} + 1)\,\overline{C}$$
$$= 1 * \overline{C}$$
$$= \overline{C}$$

C. $\overline{C}$

3. Data Structures

The binary search tree for MASSACHUSETTS is:



One child nodes are: C, H, S, S, T, U. So there are 6 of them.

D. 6

4. Data Structures

The stack is constructed using LIFO as follows:
B, BO, B, BS, B, BT, BTO, BT, B, BN, B
Therefore, the next item popped is B.

B. B

5. Boolean Algebra

$A(\overline{B + C}) + \overline{B\overline{C}} + C$
$= A(\overline{B} * \overline{C}) + \overline{B} + C + C$
$= A * \overline{B} * \overline{C} + \overline{B} + C$
$= \overline{B}(A\overline{C} + 1) + C$
$= \overline{B} + C$

B. $\overline{B} + C$

6. Boolean Algebra

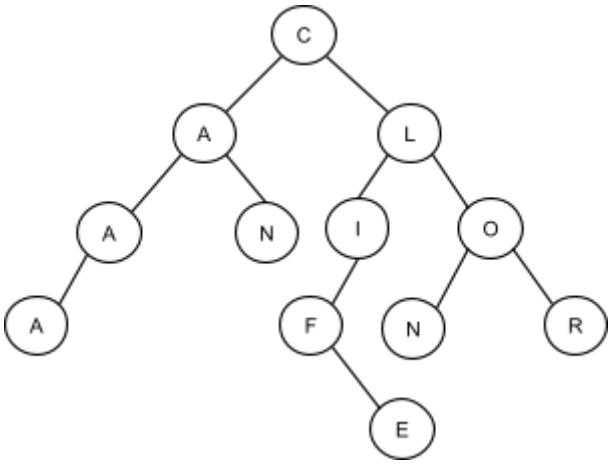
$\overline{A(\overline{B + C}) + \overline{B + \overline{C}}}$
$= \overline{A(\overline{B + C})} * (B + \overline{C})$
$= (\overline{A} + B + C) * (B + \overline{C})$
$= \overline{A}B + \overline{A} * \overline{C} + B + B\overline{C} + BC + C\overline{C}$
$= \overline{A} * \overline{C} + B$

The general ordered triples using asterisk notation (0,*,0) and (*,1,*) yields 5 ordered triples: (0,0,0), (0,1,0), (1,1,1), (0,1,1), (1,1,0)

B. 5

7. Data Structures

The binary search tree for **CALIFORNIA** is:



The internal path length is the sum of the depths of all of the nodes which is 2*1+3*2+3*3+1*4=21

D. 21

8. Data Structures The operations on an initially empty queue that is FIFO is as follows: S, SA, SAC, AC, ACR, ACRA, CRA, RA, RAM, RAME, AME, AMEN, AMENT, AMENTO, MENTO Therefore, the next item popped from the front of the queue is M.	E. M
9. FSAs The direct translation of this FSA to a regular expression is: 100*101*1. There are no other choices that are equivalent expressions to this one.	E. 100*101*1
10. FSAs and Regular Expressions Given the regular expression **`[a-ct-v][e-m|o-u]*[aeiou]?`** the words are evaluated as follows: violin - stops at n cello - valid trombone - stops at b clarinet - stops at r trumpet - stops at last t oboe - stops at o saxophone - stops at s tuba - stops at b flute - stops at f There is only 1 that is accepted.	A. 1